

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-4/T-1 CSE 409: Computer Graphics

Time: 25 minutes

Marks: 20

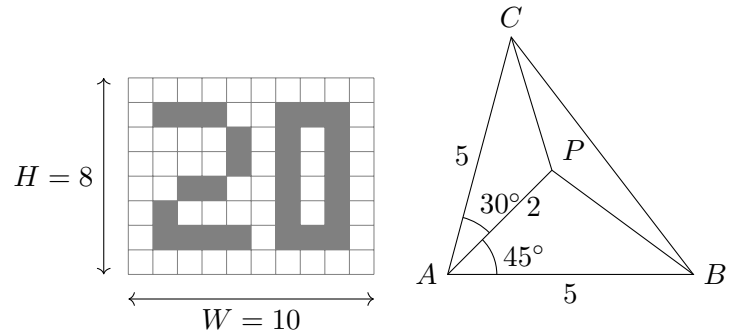
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- What is the formula for the **half-way** vector in the Blinn-Phong model? (1)
A. $\frac{\vec{L}+\vec{V}}{2}$ B. $\frac{\vec{L}+\vec{H}}{\sqrt{2}}$ C. $2(\vec{L} \cdot \vec{N})\vec{N} - \vec{L}$ D. $\frac{\vec{L}+\vec{V}}{|\vec{L}+\vec{V}|}$ ✓
- Ambient light** is an approximation of — (1)
A. **Global illumination** ✓ B. Path tracing C. Shadows D. Reflection
- For the **softest edges**, we should use spot light falloff exponent $e = ?$ (1)
A. 0 B. 2 C. 10 D. 20 ✓
- Which type of **camera preserves distances**? (1)
A. Pinhole B. Simple field C. Fisheye D. **Orthographic** ✓
- Which is **not** a feature of the **simplified pinhole camera**? (1)
A. Non-inverted image B. **Perspective distortion** ✓ C. **Depth of field effect** ✓ D. Unrealizable
- Same triangle again! But now we want to texture it with an image. The (u, v) coordinates of the vertices A, B and C are **(0,0)**, **(1,0)** and **(0.5,1)** respectively.

■ (20, 20, 20)

□ (200, 200, 200)



- What is the colour of P using nearest neighbour interpolation? (3)

Answer: From CT-3 we get :

$$\alpha = 0.5, \quad \beta = 0.207, \quad \gamma = 0.293$$

Therefore, the (u, v) coordinates of P are,

$$u = \alpha \cdot 0 + \beta \cdot 1 + \gamma \cdot 0.5 = 0.3535$$

$$v = \alpha \cdot 0 + \beta \cdot 0 + \gamma \cdot 1 = 0.293$$

We also have, $w = W - 1 = 9$ and $h = H - 1 = 7$.

$$\begin{aligned} \text{color}(P) &= T[\text{round}(u \cdot w), \text{round}(v \cdot h)] \\ &= T[\text{round}(0.3535 \cdot 9), \text{round}(0.293 \cdot 7)] \\ &= T[3, 2] = \text{rgb}(200, 200, 200) \end{aligned}$$

- What is the colour of P using bilinear interpolation? (3)

We can see that the corner colours are:

$$c_{00} = T[\lfloor u \cdot w \rfloor, \lfloor v \cdot h \rfloor] = T[3, 2] = \text{rgb}(200, 200, 200)$$

$$c_{10} = T[\lfloor u \cdot w \rfloor + 1, \lfloor v \cdot h \rfloor] = T[4, 2] = \text{rgb}(200, 200, 200)$$

$$c_{01} = T[\lfloor u \cdot w \rfloor, \lfloor v \cdot h \rfloor + 1] = T[3, 3] = \text{rgb}(20, 20, 20)$$

$$c_{11} = T[\lfloor u \cdot w \rfloor + 1, \lfloor v \cdot h \rfloor + 1] = T[4, 3] = \text{rgb}(200, 200, 200)$$

Therefore, the final colour is,

$$\begin{aligned} \text{color}(P) &= \text{lerp}(\text{lerp}(c_{00}, c_{10}, t_u), \text{lerp}(c_{01}, c_{11}, t_u), t_v) \\ &= \text{lerp}(\text{rgb}(200, 200, 200), \\ &\quad \text{lerp}(\text{rgb}(20, 20, 20), \text{rgb}(200, 200, 200), 0.1805), 0.051) \\ &= \text{lerp}(\text{rgb}(200, 200, 200), \text{rgb}(52.49, 52.49, 52.49), 0.051) \\ &\approx \text{rgb}(192.477, 192.477, 192.477) \\ &= \text{rgb}(192, 192, 192) \end{aligned}$$

Now, we can calculate the interpolation factors:

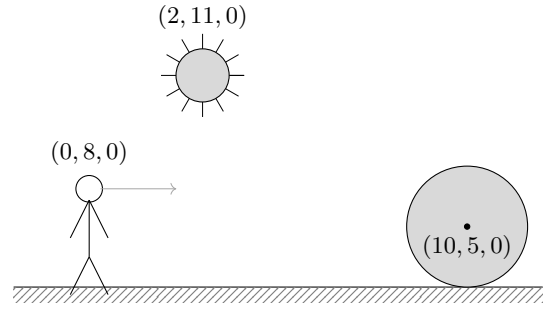
$$t_u = u \cdot w - \lfloor u \cdot w \rfloor = 0.3535 \cdot 9 - 3 = 0.1805$$

$$t_v = v \cdot h - \lfloor v \cdot h \rfloor = 0.293 \cdot 7 - 2 = 0.051$$

7. As a game developer in Iron studio, you are working in a new level of a game. There is just a strange ball at $(10, 5, 0)$ of radius 5 and a point light at $(2, 11, 0)$.

After much thought, you have decided the following parameters using the Phong model:

- Point light intensity $\mathbf{I} = (20, 20, 20)$
- Ambient light $\mathbf{I}_a = (0.1, 0.1, 0.1)$
- Ambient Coefficient $\mathbf{k}_a = (0.1, 0.1, 0.1)$
- Diffuse Coefficient $\mathbf{k}_d = (0.4, 0.4, 0.4)$
- Specular Coefficient $\mathbf{k}_s = (0.5, 0.5, 0.5)$
- Shininess $n = 10$



But your game designer says the scene **doesn't look right**. She says the ball should be **bluish gray** in shadow **$\text{rgb}(0, 10, 20)$** . She also says the material should **be less shiny**. When the character at $(0, 8, 0)$ looks straight ahead (**in the direction of x**), he should see a **gray colour $\text{rgb}(115, 125, 135)$** .

(**Hint:** Normalize RGB colours to $(0, 1)$ range.)

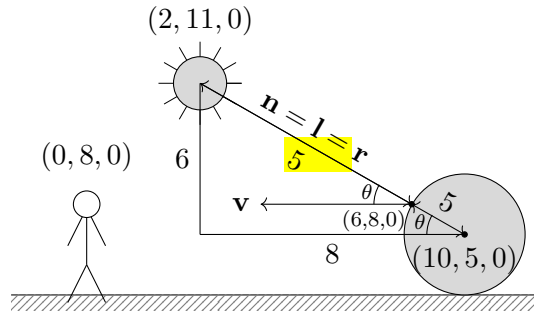
- a) As per your game designer's requirement, what should be the value of the ambient coefficient $\mathbf{k}_a$? (3)

Answer: When the ball is in shadow, only ambient light contributes to the colour. Therefore,

$$\begin{aligned} \text{colour}_{\text{shadow}} &= \mathbf{I}_a \odot \mathbf{k}_a \\ \left(0, \frac{10}{255}, \frac{20}{255}\right) &= (0.1, 0.1, 0.1) \odot \mathbf{k}_a \\ \mathbf{k}_a &= \left(0, \frac{10}{25.5}, \frac{20}{25.5}\right) = (0, 0.392, 0.784) \end{aligned}$$

- b) What should be the value of shininess n ? (3)

Answer:



Now, to calculate the final colour, **we** need to know the angles or rather the dot products $\mathbf{n} \cdot \mathbf{l}$ and $\mathbf{r} \cdot \mathbf{v}$. Since the surface is a sphere, the normal points in the direction of the light. The light is also reflected in the same direction as the normal. So, **$\mathbf{n} = \mathbf{l} = \mathbf{r}$ and $\mathbf{n} \cdot \mathbf{l} = 1$** .

The angle between $\mathbf{v}$ and $\mathbf{n}$ is the angle θ which is the angle between the sides 10 and 8, in the right triangle with sides 6, 8 and 10 between the light and the sphere center. Therefore,

$$\mathbf{v} \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{r} = \cos \theta = \frac{8}{10} = 0.8$$

Also, we need to consider the light attenuation due to distance. The distance between the light and the point on the sphere is 5.

So, now the final colour is,

$$\begin{aligned} \text{colour} &= \mathbf{I}_a \odot \mathbf{k}_a + \frac{1}{d^2} [\mathbf{I} \odot \mathbf{k}_d (\mathbf{n} \cdot \mathbf{l}) + \mathbf{I} \odot \mathbf{k}_s (\mathbf{r} \cdot \mathbf{v})^n] \\ \left(\frac{115}{255}, \frac{125}{255}, \frac{135}{255}\right) &= \left(\frac{0}{255}, \frac{10}{255}, \frac{20}{255}\right) + \frac{1}{5^2} [(20, 20, 20) \odot (0.4, 0.4, 0.4)(1) + (20, 20, 20) \odot (0.5, 0.5, 0.5)(0.8)^n] \end{aligned}$$

Solving for the n in the R component, we get,

$$\begin{aligned} \frac{115}{255} &= 0 + \frac{1}{25} [8 + 10(0.8)^n] \\ (0.8)^n &= \frac{\frac{115 \cdot 25}{255} - 8}{10} = \frac{167}{510} \\ n &= \log_{0.8} \frac{167}{510} \approx 5.003 \approx 5 \end{aligned}$$

8. For your hard work and attending the exam so early in the morning. (3)

Answer: Thanks, I suppose.